

Open Source Smart Espresso Machine

TED Talk, 2/6/26

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Who are we again?

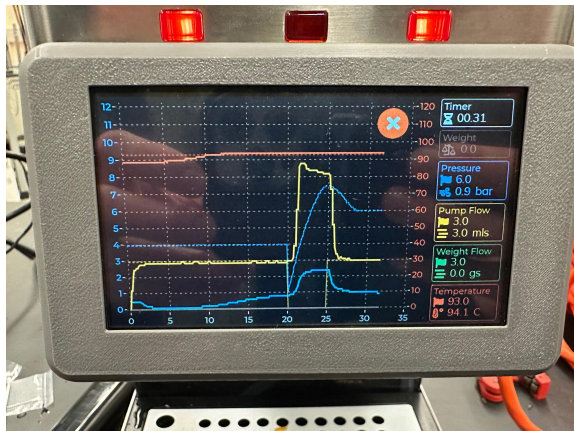
- PSU **OSSEM**
 - Open Source Espresso Machine
- Sponsored by **Healthy Buildings Research Laboratory**
- Also sponsored by **Coffee & Telesensing Laboratory**
- Functional enhancement modification for Gaggia Classic espresso machine, powered by STM32 hardware with open-source* software/firmware installations

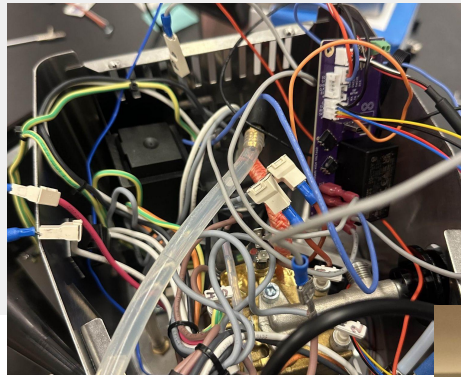
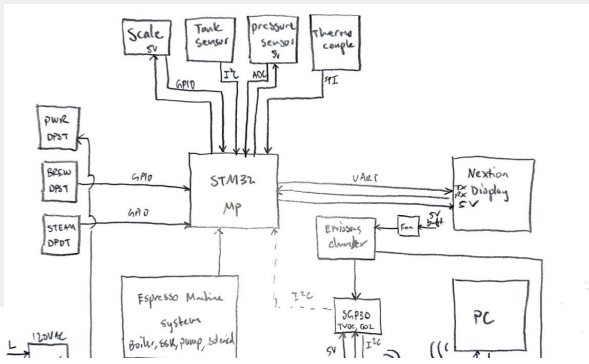
* - please don't ask ;~;



Get to the new stuff already...

- **Installed Gaggiuino hardware/microcontroller** onto our second Gaggia espresso machine
 - We can actually make coffee now (yippee)
- Split workload of microcontrollers into **two isolated devices**
 - **STM-32** inside E.M. handling machine ops, brew data logging
 - (currently) **ESP-32** "attached to" emissions chamber, driving sensors and chamber fan
- Writing and reading **lots of code**
 - Doing research on Gaggiuino/Gaggimate software
 - Microcontroller coherence (STM vs ESP) testing





etc!

What's next for the project:

- **Additional sensors** for espresso and chamber microcontrollers
 - Case temp/humid, pressure
 - pH in water chamber
- Wired + **wireless data transmission**
- **Automatic data-logging** to local, cloud storage
 - auto-snapshot for graphs

